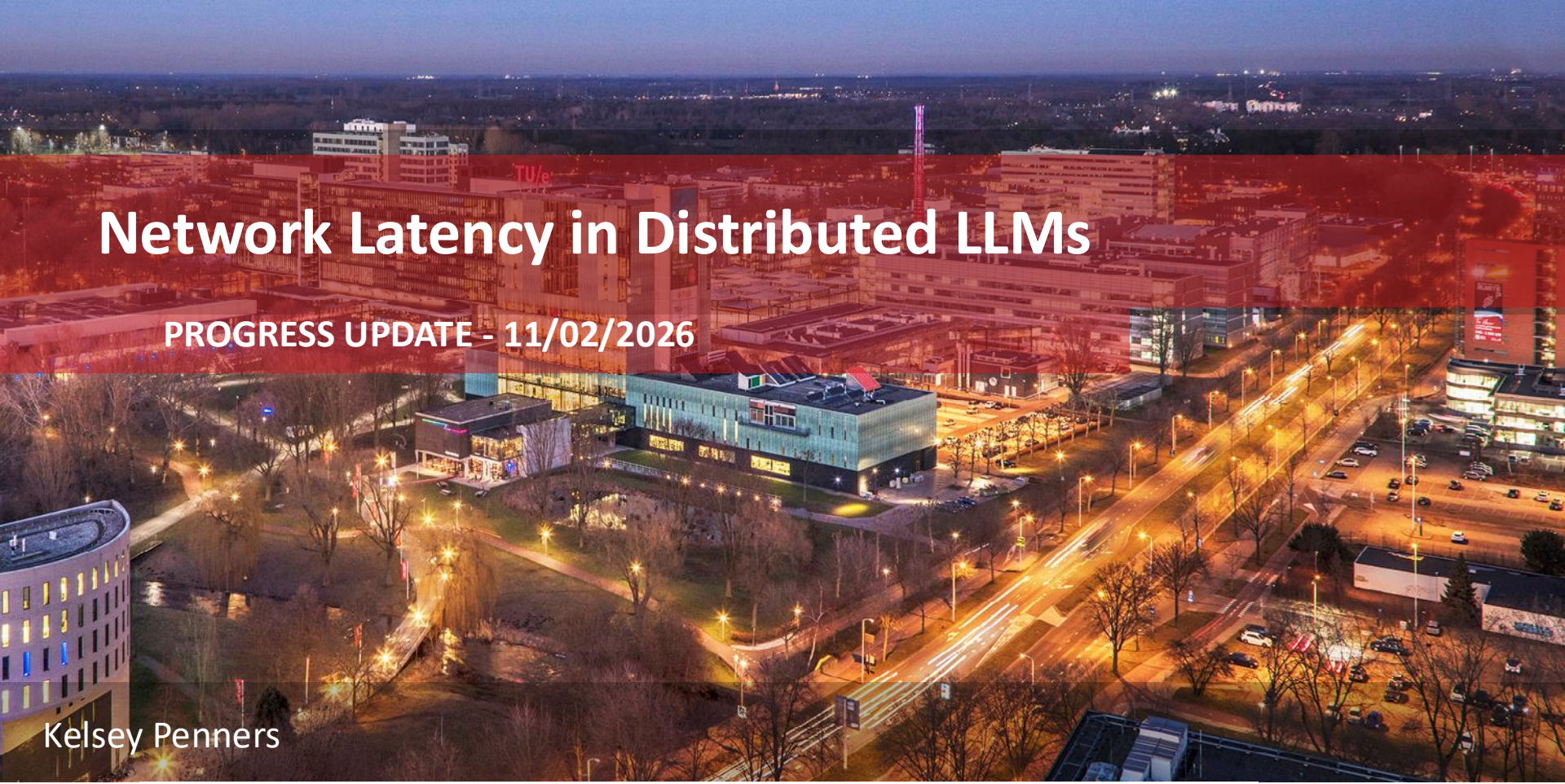


Network Latency in Distributed LLMs

PROGRESS UPDATE - 11/02/2026

Kelsey Penners

Single-GPU Baseline

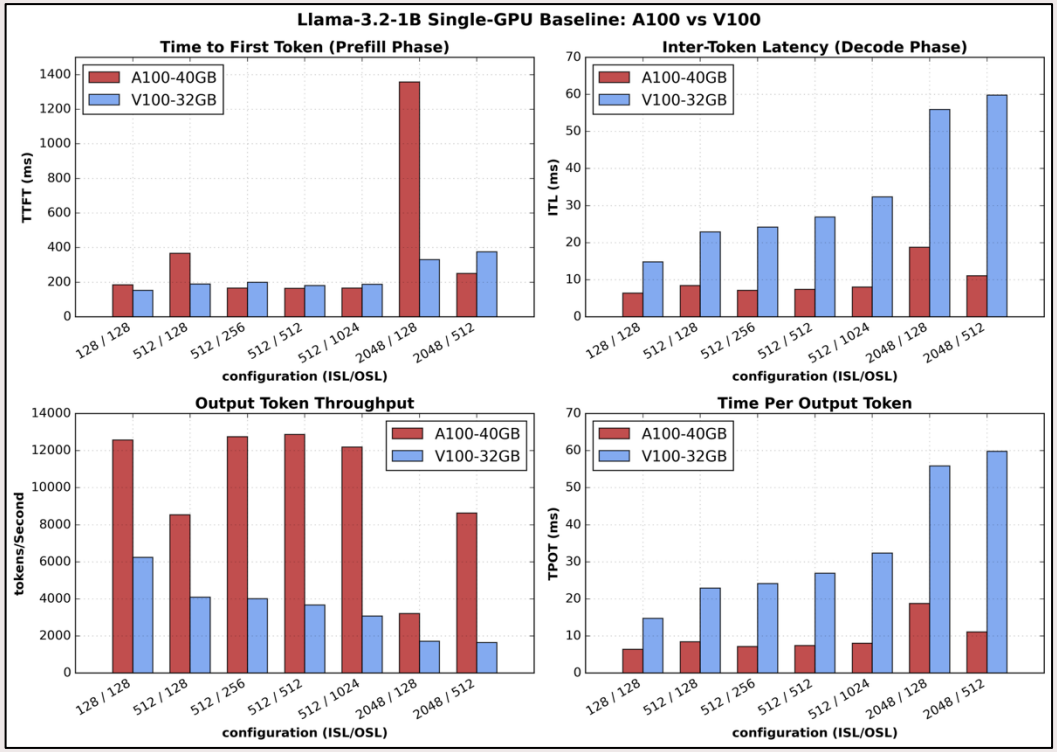
Experimental Setup:

- Model: Llama 3.2-1B – a dense LLM
- Hardware: V100-32GB & A100-40GB
- vLLM bench serve – with 100 random prompts
- Tested various configurations of ISL / OSL
  - ISL = {128, 512, 2048} tokens; OSL = {128, 256, 512, 1024} tokens

Metrics:

- TTFT (time to first token): prefill time + network latency
- ITL (inter-token latency): measures time between tokens during decode phase
- TPOT (time per output token): overall generation efficiency
- Throughput: tokens per second system capacity

Initial Performance Analysis



Metric Ranges:
A100 TTFT range: 164.3 - 1356.9 ms
V100 TTFT range: 152.4 - 376.1 ms

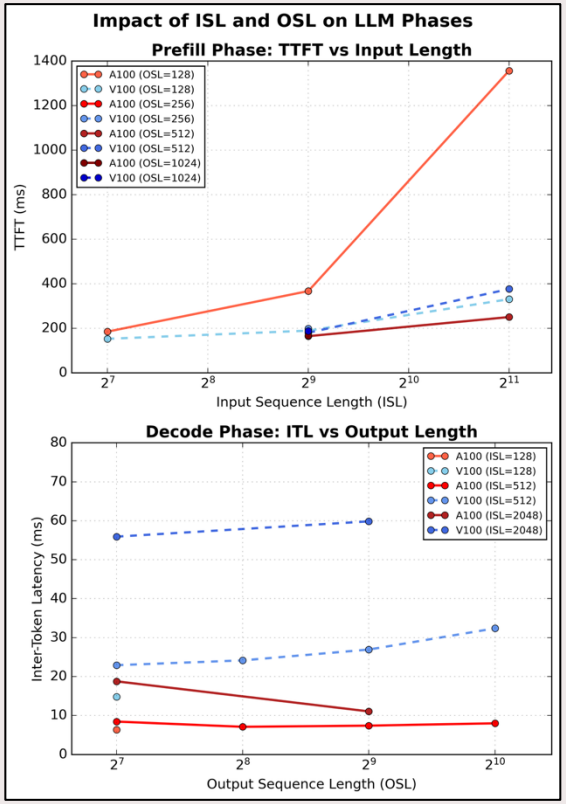
A100 ITL range: 6.4 - 18.8 ms
V100 ITL range: 14.8 - 59.8 ms

A100 throughput range: 3209 - 12873 tokens/s
V100 throughput range: 1652 - 6241 tokens/s

Observations:

- Possible anomaly in TTFT for A100
- A100 beats V100 in throughput and ITL: faster computations -> faster token generation
- Longer ISL decreases throughput on both GPUs (~ 75% reduction)
- Not much difference between ITL and TPOT

Comparison: LLM Phases Across Configs



Observations:

- Anomaly in A100 heavy load run
- ITL of A100 runs much lower and stable than that of V100 runs

Planned Experiments

- Baseline: single GPU, dense small model on shy-fec
- Measure TTFT, throughput, inter-token latency,
- Intra-node scaling: TP1 to TP 8, could compare NVLink vs PCIe (shy-fec vs funnotch), continue with same model and metrics, also try bigger models
- Inter-node scaling: use Ray to get nodes working together, run prev models, but also larger ones. Latency injection to happen here
- Heterogeneous inter-node testing: test baseline model on different systems

Next steps

Impact of ISL and OSL on Prefill and Decode Phases

